DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAMINATION
January, 2019

Probability (Ph.D. Version)

Instructions to the Student

a. Answer all six questions. Each will be graded from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your code number (**NOT YOUR NAME**) on the outside cover.

c. Keep scratch work on separate pages in the same booklet.

d. If you use a "well known" theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.

Problem 1.

Assume that X, Y are independent random variables and $X+Y$ is distributed as X. Prove that $Y \equiv 0$, $\mathbb{P}$-a-s.

Problem 2.

Let $\{X_n\}_{n\geq 1}$ be a sequence of independent random variables, uniformly distributed in $[-1,1]$. Let

$$Z_n = \frac{\sqrt{n}\sum_{k=1}^n X_k}{\sum_{k=1}^n X_k^2 + \sum_{k=1}^n X_k^3}, \quad n \in \mathbb{N}.$$

Prove that the sequence $\{Z_n\}_{n\geq 1}$ converges weakly to a normal random variable and calculate the mean and the variance of the limiting distribution.

Problem 3.

Let $\{(\xi_n, \eta_n)\}_{n\geq 1}$ be a sequence of independent and identically distributed random vectors. Assume that all ξ_n and η_n take values in $\mathbb{Z}$. Moreover, assume that

$$\mathbb{E}\xi_1 = \mathbb{E}\eta_1 = 0, \qquad \mathbb{E}\xi_1\eta_1 = c,$$

and both ξ_1 and η_1 have finite and non-zero variances.

Let X_0 and Y_0 be positive integers and let us define

$$(X_{n+1}, Y_{n+1}) = (X_n, Y_n) + (\xi_{n+1}, \eta_{n+1}).$$

Finally, let

$$\tau = \min \{n \geq 1 \ : \ X_n Y_n = 0\},$$

be the first time the random walk (X_n, Y_n) touches one of the axes.

1. Prove that $X_n Y_n - cn$ is a martingale and τ is a stopping time.
2. Let $\tau_m = \tau \wedge m$, for every $m \in \mathbb{N}$. Show that
$$\mathbb{E}\tau_m = \frac{1}{c}\left(\mathbb{E}(X_{\tau_m} Y_{\tau_m}) - X_0 Y_0\right).$$
3. Assume that $\mathbb{P}(\tau < \infty) = 1$. Determine whether it is always true that
$$\lim_{m\to\infty} \mathbb{E}(X_{\tau_m} Y_{\tau_m}) = \mathbb{E}(X_\tau Y_\tau). \tag{1}$$
4. Assume that formula (1) holds. Prove that
$$\mathbb{E}\tau = -\frac{1}{c} X_0 Y_0.$$

Problem 4.

1. Let $\{\xi_n\}_{n\geq 1} \subset L^1(\Omega, \mathcal{F}, \mathbb{P})$ and $\xi \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ be non-negative random variables such that
$$\lim_{n\to\infty} \xi_n = \xi, \quad \mathbb{P} - \text{a.s.} \ \text{ and } \ \lim_{n\to\infty} \mathbb{E}\xi_n = \mathbb{E}\xi.$$
Prove that
$$\lim_{n\to\infty} \mathbb{E}|\xi_n - \xi| = 0.$$
Hint: For every $c \in \mathbb{R}$ it holds $|c| = c + 2\max(-c, 0)$.
2. Assume that $\{X_n\}_{n\geq 1}$ and X are random variables with density functions f_{X_n} and f_X. Prove that if
$$\lim_{n\to\infty} f_{X_n}(x) = f_X(x), \qquad x \in \mathbb{R},$$
then
$$\lim_{n\to\infty} \sup_{A \in \mathcal{B}(\mathbb{R})} |\mathbb{P}(X_n \in A) - \mathbb{P}(X \in A)| = 0.$$
Hint: Use ideas from part 1.

Problem 5.

Let $\{\xi_n\}_{n\geq 1}$ be a sequence of independent and identically distributed random variables, taking values in the set of integer numbers. For every $n \in \mathbb{N}$, let

$$S_n = \sum_{k=1}^{n} \xi_k, \qquad S_0 = 0,$$

and let η_n be the number of different values taken by the sequence $S_0, S_1, \ldots, S_n$.

1. Prove that
$$\mathbb{P}\left(\eta_n = \eta_{n-1} + 1\right) = \mathbb{P}\left(S_1 \neq 0, \ldots, S_n \neq 0\right).$$

2. Prove that
$$\lim_{n\to\infty} \frac{1}{n}\mathbb{E}\eta_n = \mathbb{P}\left(S_k \neq 0,\ k \geq 1\right).$$

Problem 6.

Let $B(t)$ be a standard Brownian motion and let $Z(t) = B(t) - tB(1)$. Define

$$X(t) = (1+t)Z(t/(1+t)), \qquad t \geq 0.$$

Show that $X(t)$ is a standard Brownian motion.